DRUMPANEL · DESIGN DECISION RECORD

DrumPanel design for construction

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DOCUMENT CONTROL

Document ID	DMP-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
Source	docs/decisions/0002-design-for-construction.md @ de21d74		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Changes that make the concept physically buildable, with the reason for each, decided under Amish's pre-approval of 2026-10-03

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** accepted. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Context

STANDARDS section 18 asks for the design to be made constructable as the build plan is drawn (Amish, 2026-09-30: "fix the design assumptions to match and be physically feasible"). The TRL 2 concept listed eight components in words only. Drawing every part in `cad/src/model.py`, checking it with build123d, and sizing it in DMP-CAL-001 found the problems below. Every change keeps what DrumPanel does (purge, check, cold cut and roll flat, by hand) and its pitch. P7 changes the output from one sheet to three panels per drum; it is recorded here because the one-sheet output cannot be made, and the safety case is unchanged.

The model now runs its own checks with `python cad/src/model.py --check`: no two of the 42 components overlap, the 49 joints that must touch do touch, the shear head clears the drum at all six ring cut positions and at seven stations along the slit, and its parked position clears the posts and the drum. All pass.

Decision

Table 1. Changes made to the model, the BOM and the calculations.

#	Problem in the concept	Change made	Why this way
P1	A tilting drain stand: a drum full of water (about 240 kg) would have to be tipped	A fixed stand sloping 2.9 degrees to the bung end; the drum is filled and drained in place, never lifted full	Simpler and safer; drains to within 1.4 L (DMP-CAL-001, section 2)
P2	"Cradle and clamps" did not say what the drum sits on or how it is held	Four rollers under the rolling hoops, 380 mm apart, so the drum turns; a screw brake with a rubber pad holds it still	The hoops are the stiffest part of the body; the drum must both turn (ring cuts) and stay still (slit)
P3	The end cutter was to remove each end; a hand wheel cannot cut through the five-layer chime seam	The end cutter cuts the head out just inside the chime; the chime ring stays on the body and is cut off later with the shear head	The head plate is single thickness; the chime's inside wall acts as the die
P4	A "lever or crank-driven slitting tool" with nothing to guide it or react its force	A rotary shear head on a trolley running on an overhead rail between two posts; the upper disc is driven, the lower idles inside the drum, and a 10 mm web in the cut joins them	A gear between the discs would have to pass through the drum wall; the rail keeps the slit straight (R4)
P5	No way to start a cut through the chime rings and hoops	A 40 mm hacksaw notch through each chime ring and each hoop on the slit line before slitting	The discs cannot shear a folded seam or a 12 mm ridge
P6	A three-roll pyramid cannot drive the sheet by friction alone	A slip roll: pinch pair geared 1:1 and pressed together with 2 kN by two M16 screws, bending roll 90 mm behind on two M20 screws	Traction comes from the pinch, bending from the bending roll; crank 20 N (DMP-CAL-001, section 5)
P7	The concept rolled the whole body flat; rolling a hoop flat would stretch its crown about 8.5 %	The shear head also cuts 30 mm each side of each hoop; the body gives three panels about 233 x 1,800 mm and two hoop strips	Only flat panels go through the rolls; matches the three sheets per drum the craft already expects
P8	Rings and hoops cut with a separate tool	The shear head swivels 90 degrees on its pivot (index pin, two positions) for the six ring cuts while the crank turns the drum on its rollers	One tool, already on the rail
P9	No guards drawn	Disc guard with a front skirt 8 mm over the sheet; slip roll nip guards with 8 mm slots, lid, chain and gear guards; a cover over the end cutter wheel; all need a tool to remove	ISO 13857 slot distances met on paper (DMP-CAL-001, section 7); safety additions take the conservative option
P10	Panel ends that no slip roll can bend	The last 90 mm of each panel end is set by hand on a 10 mm bench plate with a dead-blow mallet	A slip roll cannot bend between its pinch line and its bending roll
P11	Lower roll cannot be fitted between two welded side frames	Side frames bolted to the base; the lower roll's bushes go on first, then the frames; upper and bending rolls drop down open slots closed by bolted top bridges	An assembly order that works
P12	Rail posts too close to the drum ends for the end cutter and the parked head	Posts at 712.5 mm each side of the middle, on bases across two cross members; frame 1,550 mm long	Room for the end cutter, its torque arm and the head's lower jaw
P13	Shear head reaction when cutting heads	A torque arm from the end cutter rests against a rail post	The drum turns; the cutter must not

#	Problem in the concept	Change made	Why this way
P14	Drum height under the rail fixed	The cradle's levelling feet set the drum top under the shear head; no height adjustment in the drop bar	Fewer parts; drums vary by a few millimetres

Consequences

- Cost: the estimated cost of the constructable design is USD 2,794 against the USD 3,000 value-engineering target (USD 206 under).
- R7 (four drums an hour) is not met: the notches and six ring cuts make the cradle the bottleneck at about 2.0 drums an hour.
- R5 (flatness) depends on finding the bending roll setting by trial for each batch of drum steel.
- The drawings DMP-DWG-001 and DMP-DWG-101 to 119, the concept media and the build plan DMP-BLD-001 are drawn from the changed model.